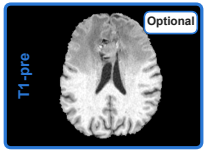
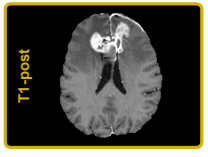
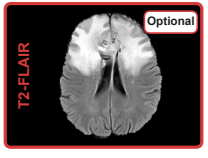
GBM Segmentation Model Architecture

6 CNNs × ~19.9 M parameters = ~119 M total

Inputs



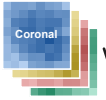


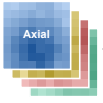


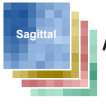


Stage 1

3 × 2.5D CNN (one per plane)







Consensus Stage 1 (gate)

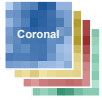
- 0 = normal brain
- 1 = abnormal brain

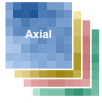
average probability > 0.30, dilated 5 voxels → gated region

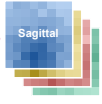
gated region

Stage 2

3 × 2.5D CNN (one per plane)





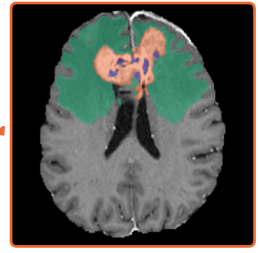


Consensus Tumor Segmentation

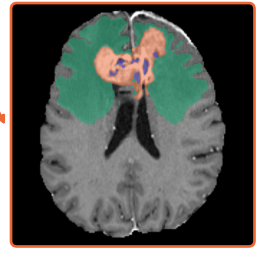
average logit → argmax

Example output (T1-post)

All four inputs



T1-post + FLAIR only (Dice 0.97)



1 necrosis
2 edema
3 enhancing tumor
4 resection cavity

T1-post is required; missing optional sequences are zero-filled. Stage 2 runs only inside the Stage 1 gated region.